Distributed Computation of the Canadian Fire Weather Index (FWI)

Decadal high-resolution gridded climate data in Chile

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Context

- Chile faces unprecedented wildfire seasons: **2017**, over 570,000 hectares burned; **2023**, over 430,000 hectares and dozens of casualties.
- Driven by 15+ years of megadrought, homogeneous flammable vegetation, and extreme days following the **30-30-30 rule** ($T > 30°C$, $RH < 30\%$, wind > 30 km/h).
- CONAF and SENAPRED operate on the **Canadian Fire Weather Index (FWI) System**: six interdependent sub-indices (FFMC, DMC, DC, ISI, BUI, FWI) that estimate fuel moisture and expected fire intensity from daily weather.

The problem we want to address

- Today the FWI is computed **only** at automatic weather stations. Chile's network is sparse, leaving **blind spots** in inner valleys and rural forest areas — where recent megafires started.
- Covering the country with an interpolated 5 km climate grid means computing $\sim$16,000 days $\times$ $\sim$20,000 cells $\approx \mathbf{3.2 \times 10^8}$ states.
- The model is **recursive**: day t depends on day $t-1$, so it cannot simply be split by date.

$$FFMC_t = f(FFMC_{t-1},\ T_t,\ RH_t,\ W_t,\ P_t)$$

- A single machine fails: out-of-memory on 44-year matrices, fragmented NetCDF reads, and 30–50 hours per sequential run.

Relevance

- **Operational:** a nationwide daily fire danger map instead of a handful of isolated points, directly useful for prevention and resource allocation.
- **Scientific:** 44 years of reconstructed FWI enable trend analysis of fire danger under the megadrought.
- **Technical:** the case is a clean instance of a recurring Big Data challenge — distributing **stateful, time-recursive models** over massive gridded data without saturating the network.

Research question

How to structure and partition a massive multi-decadal climate dataset on a distributed cluster (Apache Spark) to compute time-recursive models efficiently?

Data we plan to work with

Provider	CR2, Universidad de Chile (`vismet.cr2.cl`)
Product	**CR2MET v2.5** — daily precipitation and max/min temperature
Format	NetCDF-4, arrays `[time, lat, lon]`
Resolution	0.05° (≈ 5 km), continental Chile
Coverage	1979–2023, 44 years of continuous daily records
Volume	Tens of GB compressed; $>$ 320 million states to compute
Variety	Raster grids + station tables for validation + land cover

Processed with **Apache Spark** over **Apache Parquet**, partitioned by spatial cell so each executor holds the full time series it needs.